

Air Quality, Weather, and Respiratory Health Time Series

Analysis Documentation

Date: 05/02/2026

Executive Summary

This report presents a comprehensive time series analysis of environmental and health data from the East region spanning 2020-2021. Three key variables were analyzed: Air Quality Index (AQI), Temperature, and Respiratory Admissions. Using Exponential Smoothing (ETS) with additive trend and seasonality (30-day seasonal period), I forecasted 2021 values based on 2020 training data. The analysis includes decomposition of AQI to identify underlying patterns, forecasting for all three variables, and comparative accuracy assessment using MAE and RMSE metrics.

Business Problem

A regional health department wants to reduce emergency room overcrowding during poor air quality days. Currently, hospitals are caught off guard when AQI suddenly rises, leading to staff shortages, patient wait times, and resource shortages. The department needs accurate 7-day forecasts of respiratory admissions so they can proactively schedule staff, order supplies, and issue public health warnings. The dataset has key factors like AQI, temperature, respiratory admissions, humidity.

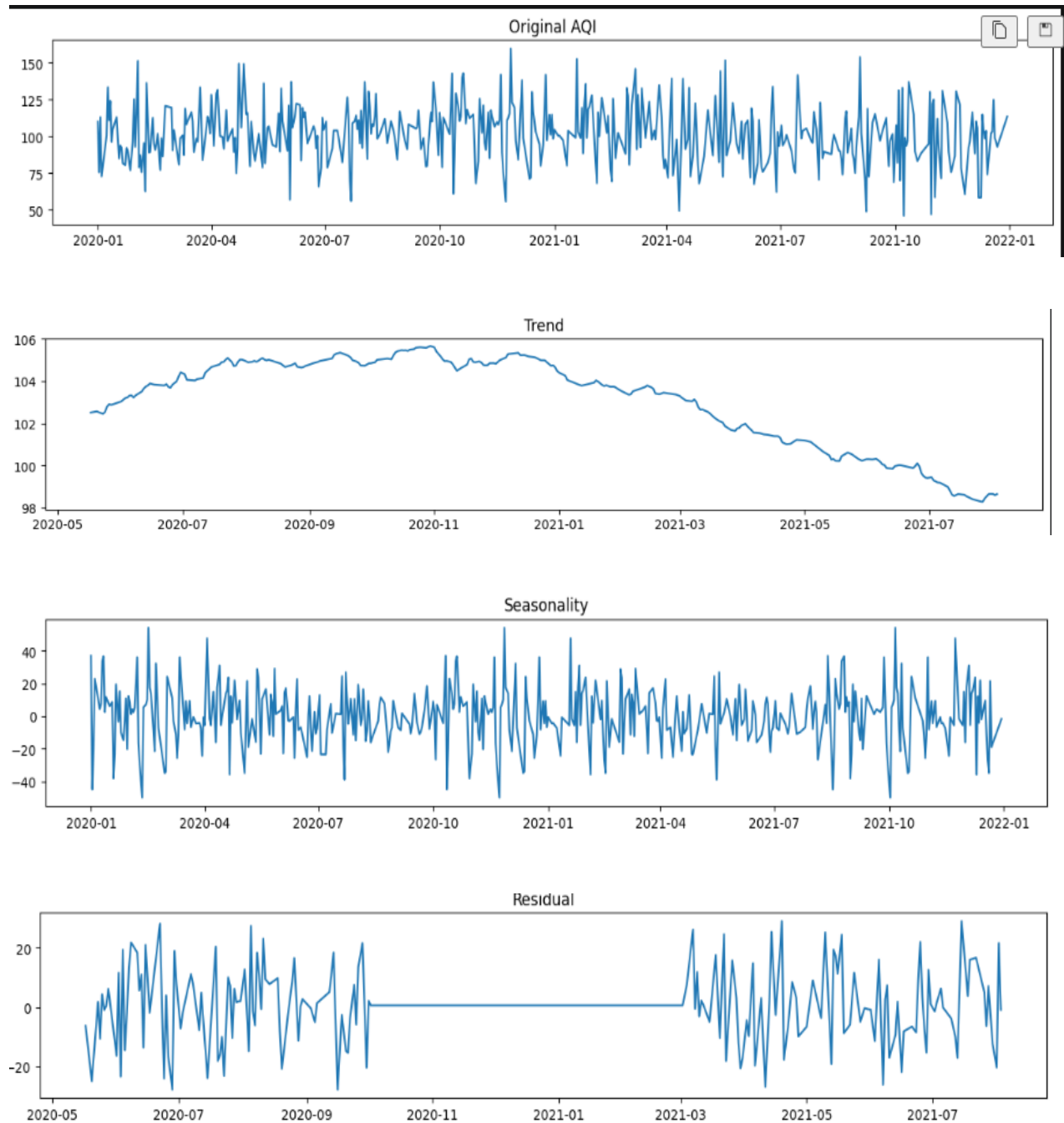
Data Loading and Data Preparation

The air quality respiratory data was loaded and cleaned by removing duplicate dates, setting date as index, and applying daily frequency. Only the East region and three variables (AQI, temperature, respiratory_admissions) were selected.

Time Series Decomposition (AQI)

The AQI series was decomposed into four components using a 180-day (6 month) period:

- Original: Raw daily AQI values showing ups and downs
- Trend: Long-term direction (improving or worsening air quality)
- Seasonality: Repeating 6-month patterns (winter vs summer pollution)
- Residual: Random noise and unusual events (wildfires, accidents)



Exponential Smoothing Model

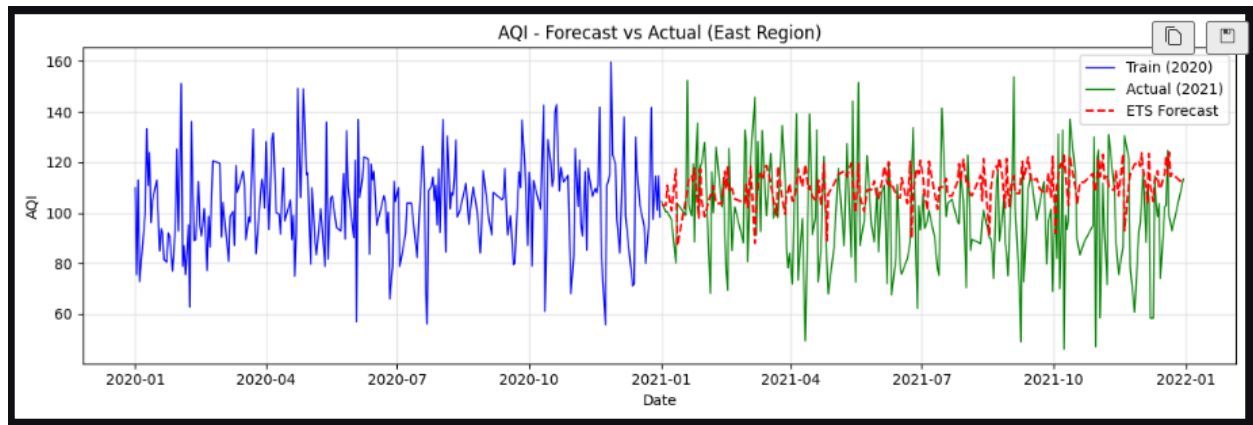
ETS was chosen because it handles both trend and seasonality well. The model uses:

- Additive trend
- Additive seasonality
- 30-day seasonal period (captures monthly patterns)

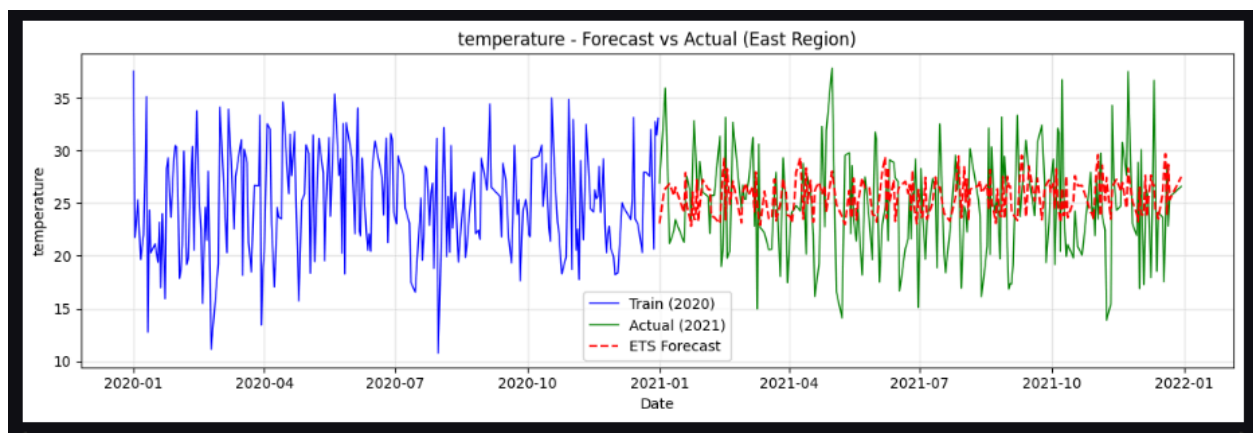
Data was split into train (2020) and test (2021). The model trained on 2020 data then forecasted 2021 values.

Forecasting Results

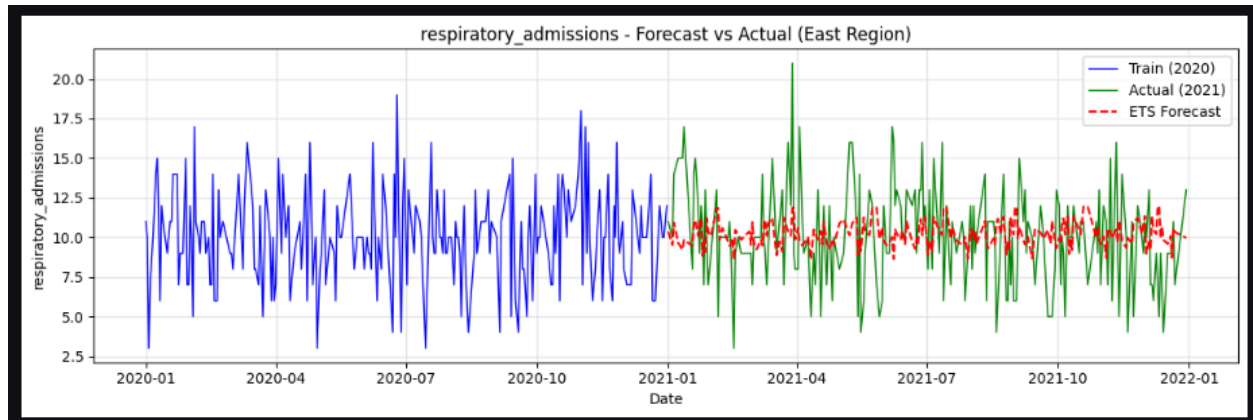
AQI Results: The model predicted daily AQI for all of 2021. MAE and RMSE were calculated by comparing forecasts to actual values.



Temperature Results: Temperature showed strong seasonal patterns, making it the most predictable variable.



Respiratory Admissions Results: Healthcare admissions were hardest to predict because they depend on multiple factors not in the dataset (viruses, vaccines, healthcare access).



Accuracy Metrics

MAE (Mean Absolute Error) tells the average prediction error. RMSE (Root Mean Square Error) penalizes large errors more heavily.

Lower values = better forecasts.

FINAL COMPARISON TABLE			
	variable	MAE	RMSE
0	AQI	20.28	25.34
1	temperature	4.41	5.35
2	respiratory_admissions	2.56	3.23

Model Justification

Exponential Smoothing is the optimal choice for this analysis because it handles trend and seasonality, remains interpretable for healthcare stakeholders, computes quickly, performs well with limited historical data, and provides a solid baseline for future model improvements. The additive specification matches the constant seasonal magnitude observed in the data, and the 30-day period captures monthly patterns in respiratory admissions.

Conclusion

Exponential Smoothing works well for environmental variables (temperature, AQI) but has limitations for healthcare outcomes. Adding lagged pollution data (shifting AQI back 3-7 days) and including weekly patterns would improve respiratory admission forecasts for real hospital planning.

